



Interactive Video Search with Multi-modal LLM Video Captioning

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Abstract. Cross-modal representation learning is essential for interactive text-to-video search tasks. However, the representation learning is limited by the size and quality of video-caption pairs. To improve the search accuracy, we propose to enlarge the size of available video-caption pairs by leveraging multi-model LLM on video captioning. Specifically, we use LLM to generate video captions for a large video collection (i.e., WebVid dataset) and use the generated video-caption pairs to pre-train a text-to-video search model. Additionally, we use LLM to generate fine-grained captions for test video collections to enable text-to-caption retrieval. Furthermore, we build a semantic overview of the retrieved rank list based on the detailed captions in our interactive video retrieval system which act as hints for user to refine their query. Experimental results show that the generated captions are effective in improving the search accuracy of both AVS and T-KIS tasks on the TRECVID datasets.

Keywords: Interactive Video Retrieval · Multi-modal LLM · Video Captioning

1 Introduction

Utilizing large pre-trained cross-modal representation models as interactive search engines have been proven to be effective. This trend is predominantly evident in the annual interactive video search competition of Video Browsing Shutdown¹ (VBS) [4, 8, 16], where for the past three years, the winners [1, 5, 14], all had applied CLIP [12], or variations of it, as their back-end. However, the effectiveness of cross-modal representation learning usually relies on the size and the quality of the training data. Although there is a large amount of available

¹ <https://videobrowsershowdown.org/>.

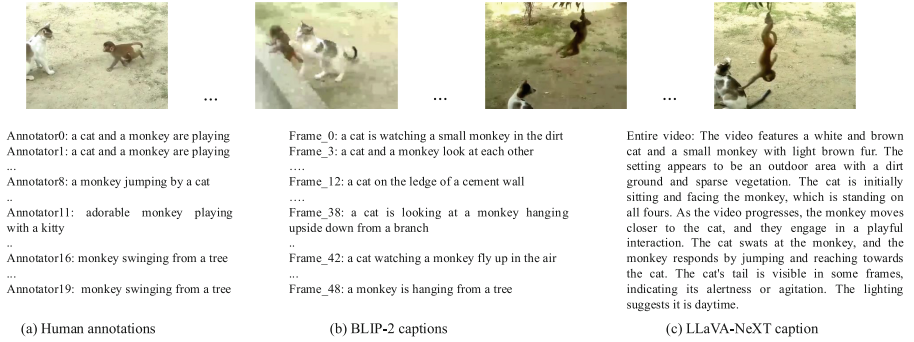


Fig. 1. Video caption comparison between the human annotators, BLIP-2 model, and LLaVA-Next model.

text-image data, such as LAION-5B [15], the size of the text-video pairs is limited. Thus, it hinders the development of the text-video search models [19]. Especially those models targeting the ad-hoc video search task have a higher chance of encountering out-of-vocabulary problems because of the open-vocabulary setting.

One straightforward attempt to address this challenge is by enlarging the size of existing text-to-video pairs for training. For example, millions of text-video pairs are introduced in the WebVid2M dataset [2]. Specifically, this dataset has 2.5M video clips, and each of them is associated with a manually annotated caption. Knowing that text is limited in expressing visual concepts, there are also attempts to transform text query into visual query [10], by generating visual examples using diffusion models [10] for query-by-images retrieval.

Following the same line of approaches, we also enlarge the size of text-video pairs for pre-training our model in [18]. However, different from previous works, which manually annotate the video content [2] or extract OCR from the videos [11] as text data, we leverage the multi-modal LLM to automatically generate video captions to provide a rich textual description of the videos. Specifically, we have utilized two cross-modal LLMs, i.e., BLIP-2 [6] and LLaVA-NeXT-Video [21], for text-to-video retrieval model pre-training and text-to-caption retrieval respectively. BLIP-2 is used to generate about 7 million captions for around 1.44 million videos in the WebVid dataset [2], and the generated captions are subsequently used to pre-train our previous cross-modal search engine (i.e., ITV [9]). In addition, we use BLIP-2 and LLaVA-NeXT-Video to generate captions for every video in the test video collections (e.g., V3C1 [3]) to enable text-to-caption retrieval. Figure 1 contrasts the captions for a video in the MSRVT dataset [20]: (a) human annotations, (b) BLIP-2 captions, (c) LLaVA-NeXT caption. The human annotations tend to describe the big picture of a video with a short sentence, such as “a cat and monkey are playing”, and there is much redundancy between different annotators. The captions generated by BLIP-2 are also short sentences but with more details (e.g., “on the ledge of a cement wall”,

and “hanging upside down from a branch”). In contrast, the LLaVA-Next model describes the video sequence in a paragraph with more fine-grained details, e.g., the color of the cat and monkey, the time and the location.

We have conducted ad-hoc video search (AVS) and textual known-item search (T-KIS) on the TRECVID datasets to validate the effectiveness of the pre-training and text-to-caption retrieval. The results show that improved retrieval performance is obtained by the multi-modal LLM video captioning. We also perform a fusion of the pre-trained text-to-video retrieval and the text-to-caption retrieval, which obtained a better retrieval rank on the T-KIS queries.

2 Video Captions for Pre-training and Retrieval

Given a video v , it is sent to the BLIP-2 model or LLaVA-video-Next-7B model to obtain a video caption c . Specifically, c is a list of frame-level captions in BLIP-2 and is a paragraph caption in LLaVA-Next, as shown in Fig. 1. The prompt used in the LLaVA-Next model is *“Please provide a short and concise description of the video, focusing on the main subjects, their actions, objects in the video, the location of the video, and the time of day.”* and the number of frames to process is set to 8.

The 7 million caption-video pairs obtained from captioning WebVid [2] are used to pre-train the ITV [18] with a new multi-word concept bank and new visual/textual features introduced in [19]. After pre-training the generated data and fine-tuning the ITV model on three video caption datasets [7, 17, 20], the improved ITV (IITV) is used as our text-to-video (t2v) search engine in our retrieval system. For an input query q , a video rank list $R_{t2v} = IITV(q)$ is produced by the IITV.

We also equip the search system with text-to-caption (t2c) retrieval by using the generated captions of test videos. Specifically, we use Sentence Transformer [13] (i.e., all-mpnet-base-v2 version) to encode the text query q and the generated video captions c . The rank list R_{t2c} is obtained by performing cosine similarity between $ST(q)$ and $ST(c)$.

We further perform late fusion between the rank lists from text-to-video and text-to-caption retrieval to get the resulting R_{fusion} . Empirically, video embedding is a more comprehensive representation of a video than a video caption focusing on specific viewpoints. Hence, we use a higher weight for text-to-video retrieval, where 0.8 is assigned for it versus 0.2 for text-to-caption retrieval.

3 Concept Analysis

We also capitalize on the generated captions of the test videos to provide a semantic overview of the top- k (where $k = 1000$) retrieved videos. This semantic overview provides a glance of concept distribution. The user can judge from the overview whether the retrieval result is satisfactory without the need to painstakingly browse the videos. The concept distribution also provides clues for user to reformulate the search query.

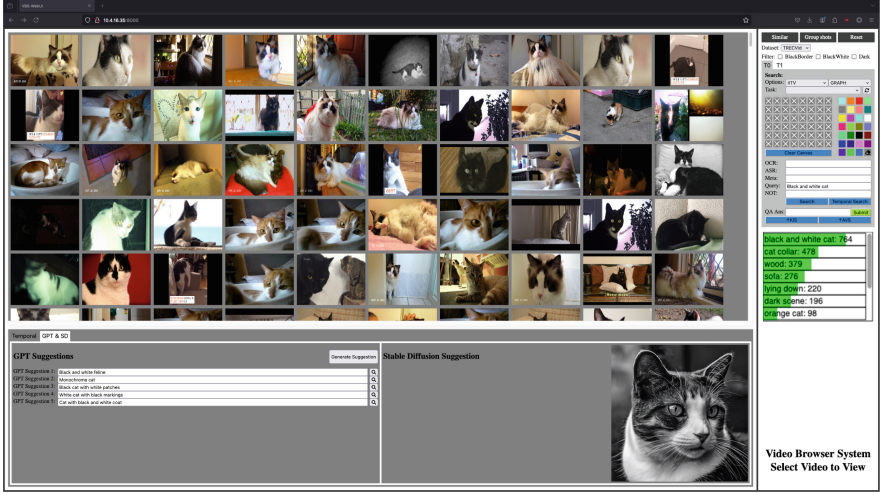


Fig. 2. User interface. The search query is “black and white cat”. The horizontal histogram on the right side illustrates the concept distribution of the current search results.

Given a rank list of k videos $R = [v_1, v_2, \dots, v_k]$, we extract n -gram words from the caption c of each video to form a concept bank, where $n = \{1, 2, 3, 4\}$. The n -gram words in the bank are sorted based on their appearance frequency and then presented to the user.

Figure 2 shows the interface of our video search system. For the input query “black and white cat”, the search engine returns a rank list of 1,000 videos, where their keyframes are displayed on the panel for browsing. The semantic overview of the current rank list is shown on the right-hand side with green bar and number showing the frequency. Users can scroll down the list to see more concepts. For this example, besides the concept *black and white cat*, the analytics indicates that most of the cats on the rank list are *lying down* and *sofa* is the frequently place they stay. The user may start browsing and hunting for the videos if the search target is a cat lying on a sofa. The analytics also provide guidance on how to refine the query. The user can provide details to supplement the original query, for example, “a black and white cat walking in the garden”, if the search target is not about a cat staying indoors.

4 Experiments

4.1 Semantic Overview of the V3C1 Dataset

The wordclouds of the top 100 frequently appear concepts extracted from the captions generated by the LLaVA-Next and BLIP-2 for the videos on V3C1 dataset is shown in Fig. 3, along with the wordcloud of the concept bank of our system. A wide spectrum of concepts are covered by the generated captions,



Fig. 3. Wordclouds of the top 100 concepts generated from the V3C1 dataset using LLaVA captions (left), and BLIP-2 captions (centre). The wordcloud of the concept bank of our search engine (IITV) is shown on the right.

including who (people), what (objects), where (e.g., *dark room* and *street*), when (e.g., *day time* and *morning*), how (e.g., *riding*, *smiling* and *camera panning*), showing the quality of captions. In total, there are 130,257 and 530,488 number of concepts from BLIP-2 and LLaVA captions, respectively. Compared to the vocabulary size of 14,528 in our system, the generated captions are able to provide more and diverse concepts, especially the LLaVA-Next model. For example, *dressed casual attire*, *dimly lit*, and *late afternoon*, are fine-grained and new concepts introduced by LLaVA-Next model.

Table 1. AVS Performance comparison of the widely used cross-modal models and our methods with and without using generated captions for pre-training.

	IACC.3			V3C1			V3C2		
Method	tv16	tv17	tv18	tv19	tv20	tv21	tv22	tv23	mean
CLIP [12]	0.182	0.217	0.089	0.117	0.128	0.178	0.124	0.109	0.143
BLIP-2 [6]	0.213	0.226	0.168	0.199	0.222	0.273	0.164	0.203	0.209
IITV w/o pre-training	0.262	0.321	0.158	0.248	0.346	0.369	0.215	0.280	0.275
IITV w/ pre-training	0.280	0.349	0.165	0.242	0.352	0.365	0.235	0.295	0.285

4.2 Ad-Hoc Video Search

We evaluate the proposed pre-training IITV model on the AVS task. In Table 1, we compare the xinfAP performance with several text-to-vision retrieval models. Specifically, CLIP [12] and BLIP2 [6] are two widely used search engines in interactive video search systems [1, 16]. We also include IITV with and without using the generated captions for pre-training in the comparison. Except for the query set tv18, IITV outperforms CLIP and BLIP-2 across seven-year query sets over three datasets. In addition, the IITV with pre-training outperforms the IITV without pre-training on most query sets, which verified the benefit of using generated captions on pre-training.

4.3 Textual Known-Item Search

We verify the effectiveness of the proposed text-to-caption (t2c) retrieval on the T-KIS task. Table 2 compares the median rank of six years of T-KIS query sets (VBS2019-2024) between our method and BLIP-2 model [6]. Specifically, its t2v result is obtained by using BLIP-2 video and text encoders, and the t2c result is obtained by using BLIP2 model for video frame captioning. T2v search significantly outperforms t2c search by having Med ranks of 23.5 and 9.92. With a more fine-grained level of video caption generated by LLaVA-NeXT, the average median rank of the t2c search is dramatically improved from over 182k to 2487.83. Moreover, the fusion with t2v search obtains the best average median across six query sets. We found that the improvement boost comes from the large variant retrieved rank lists obtained by t2v and t2c searches. For example, for the query-VBS2020-t3 finding a video of “*driving along a bike path (from the view of the bike rider), train track on the right, past a train stop, towards a crossing with a street. A path to an under path to the platform branches off on the right, and a bike is parked there. A group of bikers is waiting in front of the street crossing*”, t2v methods tend to rank videos of a train station with a bike path higher. In contrast, t2c methods tend to rank videos of a group of bikers at the top. For this query, t2v and t2c methods only manage to rank the target video at 2,798 and 3475, respectively. However, when fusing them together, the rank of the ground truth video is elevated to 70 because videos of both train station with bike path and a group of bikers have higher scores on both rank lists.

Table 2. Median rank comparison of text-to-vision (t2v) search, text-to-caption (t2c) search and their fusion on six years of T-KIS query sets.

Search model	Method	VBS2019	VBS2020	VBS2021	VBS2022	VBS2023	VBS2024	Average
t2v	BLIP-2 [6]	1.5	4.5	3.5	8	114	9.5	23.5
	Ours	8	2.5	3.5	12	19	14.5	9.92
t2c	BLIP-2 [6]	59,315	173,399	119,719	349,826.5	88,120	303,008.5	182,231.25
	Ours	20	17.5	58.5	482	542	13,807	2487.83
t2v+t2c	BLIP-2 [6]	8	2	2.5	14	31	26.5	14
	Ours	4.5	1.5	1.5	2	17	2	4.75

5 Conclusion

In this paper, we have studied two ways of utilizing the multi-modal LLMs generated captions for interactive video retrieval. The experimental results on AVS and T-KIS tasks show that it can benefit both the cross-modal model pre-training and video retrieval performance. Text-to-vision retrieval significantly exceeds text-to-caption retrieval on textual KIS. However, with more fine-grained captions obtained from the LLaVA-Next model, the performance gap between them is narrow. Moreover, the linear fusion of them has achieved the best average

median rank. We have also developed a semantic overview of the current retrieved rank list based on the generated captions to help users find the correct answers more quickly.

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